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Full Length Article



Design and fabrication of a new chest compression device with a specialized piston and bayonet lock mechanism

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ABSTRACT

Cardiopulmonary Resuscitation (CPR) is a medical intervention to address Cardiac Arrest and effective chest compressions that increase blood flow to vital organs are critical for CPR to be successful. According to the American Heart Association (AHA) guidelines, manual CPR is recommended. However, many CPR devices have recently been developed to achieve more consistent chest compressions. In this study, a portable, small, and compact mechanical CPR device with a unique piston mechanism consisting of a double slider crank and dynamic conrod, unlike the piston mechanisms of existing devices, has been designed and manufactured. For the device called CardiS to be integrated into the patient quickly and effortlessly, on the other hand, an automatic bayonet lock system with radial and axial locking functions has been developed. Thanks to this lock system, which does not require any alignment and provides locking with lower force, fast installation, a crucial parameter for CPR devices, can be easily performed. This device has been compared with the LUCAS II device regarding standard CPR parameters and compression timings using a special CPR manikin called CPR Lilly Pro+. The results indicate that both devices successfully met the standard CPR parameters specified in the AHA guidelines, but the CardiS provided a more consistent compression depth. Furthermore, the average return time of blood to the heart has been 400.4 ms for CardiS and 299.7 ms for the LUCAS II device. This longer return time allows the heart to fill better and pump more effectively.

1. Introduction

Cardiopulmonary Resuscitation (CPR) is the first aid intervention method to provide blood circulation to vital organs (brain) in cases where the heart cannot perform its blood-pumping function (Cardiac Arrest) [1]. The main intervention in CPR is chest compression, also called cardiac massage. Chest compression is generally the process of compressing the heart between the spine and sternum by applying pressure on the sternum with both hands. In this way, blood in the heart can be transmitted to vital organs, and infarction (tissue death) in these organs can be prevented. In addition, the heart can reach a shockable rhythm and the possibility of the heart restarting again can be increased [2].

Although the mechanism of action of chest compression is not fully known, two theories are accepted today: Cardiac Pump Theory [3] and Thoracic Pump Theory [4]. In the cardiac pump theory, as the heart is compressed between the spine (vertebral column) and the sternum (breastbone), the internal pressure of the ventricles increases. With this increased pressure, blood is pumped to the aorta and lungs. In the thoracic pump theory, when the pressure is applied, the pressure in the heart cavity, lung cavity, and chest cavity increases. With this increased pressure, blood is pumped to vital organs. Thus, while cardiac pump theory focuses on improving blood circulation by acting directly on the heart, thoracic pump theory attempts to increase blood circulation by optimizing changes in rib cage volume. The fundamental difference between the two theories lies in the Cardiac Pump Theory's consideration of all of the heart's valves (Tricuspid, Pulmonic, Mitral, and Aortic) as functional, whereas in the Thoracic Pump Theory, none of the heart's valves are considered functional.

The American Heart Association (AHA) and the European Resuscitation Council (ERC), are the international guidelines where there are the necessary conditions for providing high-quality CPR to adults. These guidelines recommend chest compressions 5 to 6 cm deep and at a rate

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Nomenclature

Symbols

w	Instantaneous crank angular velocity
θ_i	Crank angle at the start of compression
N_i	The initial velocity of the motor and flywheel
k_s	The friction coefficient of the piston bearing
F_L	Force acting on the conrod
w_{avg}	Average angular velocity of the crank
E_G	Gear efficiency
f	Compression frequency
F_P	The force applied by the piston to the patient chest
f_s	Friction force on the piston
F_{set}	Piston preload
I_{Fl}	The flywheel inertia
K_1	Flexibility constant of the patient's chest
K_2	Flexibility constant of the static structure
K_3	Equivalent spring constant of decompression springs
L	Conrod length
m_1	The mass modeling of the patient's chest
m_2	The mass representing the static structure
D_1	The damper constant of the patient's chest
n	The gear reduction ratio
O	Center of the double slider-crank mechanism
P	Piston
R	The radius of the crank
S	Piston stroke
w_i	Initial angular velocity of the crank
x	Instantaneous piston stroke
x_1	Instantaneous compression depth
x_2	The stretching amount of K_2 spring
θ_{max}	Maximum crank angle
θ	Instantaneous crank angle
α	Instantaneous angular acceleration
τ_{Cl}	Electromagnetic clutch nominal torque

Abbreviations

TTL	Time to live
PCB	Printed Circuit Board
CPR	Cardiopulmonary Resuscitation
AHA	American Heart Association
PWM	Pulse Width Modulation
RCTs	Randomized Clinical Trials
ERC	European Resuscitation Council
CardiS	Cardiac Stimulator

of at least 100 per minute (up to 120). In addition, according to these guidelines, the patient's chest should be allowed to fully recoil during compression and the chest compression fraction, which is the ratio of the time spent on compression to the entire resuscitation time, should be at least 80 % [5–7]. The conditions presented show that a high-quality manual CPR requires excessive force. Studies also confirm that the provider's effective chest compression ability decreases due to fatigue after a certain period [8,9]. Therefore, the AHA recommends changing the rescuer every two minutes during CPR, if possible. In addition, chest compression efficiency is adversely affected as body and arm position cannot be maintained during the transport of the patient or in a moving ambulance [10]. Many mechanical CPR devices have been developed to overcome the stated challenges and provide high-quality chest compressions with limited human interaction [11].

In this study, a novel mechanical CPR device called CardiS (Cardiac Stimulator), featuring a specialized piston mechanism comprising a double-slider crank and a dynamic conrod, was designed, manufactured, and tested on a CPR manikin whether it provides standard CPR parameters. In addition to having a patented piston mechanism [12], the developed device has been intended to contribute to the existing literature with its features such as being portable and small, offering adjustable compression depth, and ensuring stable operation. For this purpose, it has been aimed to be a successful alternative to existing CPR devices, which are not widely used due to their low success rate and high cost.

1.1. Literature review

Mechanical CPR devices are machines that provide effective and uninterrupted chest compressions by CPR guidelines to ensure sufficient blood flow to the heart and other vital organs during cardiac arrest [13]. These devices exhibit differences from one another in terms of their operating mechanisms, energy consumption, and mechanical-electronic properties. Currently used automatic CPR devices are divided into two basic groups: piston-driven CPR devices and load-distributing band CPR devices [14]. In addition to these two basic groups, devices that apply different working mechanisms and different CPR techniques such as active compression-decompression devices are also used [15]. Piston-driven CPR devices are based on Cardiac Pump Theory and apply pressure to the sternum using a mechanical piston [3]. Load-distributing band CPR devices are based on the Thoracic Pump Theory and use a pneumatic band that wraps around the chest [4]. The first of the automated CPR devices introduced to clinical and preclinical study was the Thumper device, which was developed by the University of Michigan in the mid-1970s and uses a compressed air-driven piston [16]. In this device, which is the first example of a piston-driven CPR device, the compression ratio, and depth are adjusted manually with scales. Many different piston-driven CPR devices such as Life-Stat, LUCAS, and Corpuls have been developed from the Thumper device to the present [17]. Like the Thumper device, the Life-Stat device consists of a backplate attached to a single arm and operates pneumatically [17]. The LUCAS device is a fully pneumatic-driven device consisting of a pneumatic cylinder mounted on two arms attached to a rigid backplate [18]. The force is transferred to the chest through a rubber connected to the pneumatic cylinder and chest compression is provided. Due to the laborious use of gas cylinders and high-pressure air consumption, the LUCAS II, an all-electric device, was developed [19]. Currently one of the most widely used chest compression devices in clinical practice, the LUCAS II uses a screw piston mechanism to compress the patient's chest wall and is powered by a motorized electrical power source. As in LUCAS, the force transmitted to the patient's chest is balanced by the double arm. The Corpuls device utilizes a screw piston mechanism just like the LUCAS II device. Compression is generated by a single, flexible, adaptable arm locked onto a backplate placed underneath the patient. The device provides an adjustable compression frequency ranging from 80 to 120 compressions per minute and a 2–6 cm compression depth [20]. The AutoPulse device, which is the only one of its kind with a load-distributing band, includes a wide load-distribution band that cyclically inflates and deflates around its circumference to provide chest compression. This load-distribution band is equipped with a single device that houses the control computer, rechargeable battery, and motors. Distinguishing itself from other compression devices, it delivers chest compressions at a constant rate of 80 per minute, reaching a depth of approximately 20–30 % of the chest circumference [21]. Although these developed devices can reach the standard CPR parameters recommended by the AHA, the survival rates obtained with these devices could not surpass the average manual CPR in many randomized clinical and observational studies [22–24]. Therefore, the AHA currently does not

recommend the routine use of mechanical CPR devices. However, there are situations (diagnostic imaging, organ transplantation, ambulance transport, etc.) where the use of mechanical CPR devices may be the only option, and manual CPR may be difficult or impossible to perform effectively.

Some studies in the literature using CPR devices with different piston mechanisms can be explained as follows. For instance, Alam et al. [25] developed a low-cost CPR device that satisfies the requirements of compression frequency (100–120/min) and maximum compression depth (6 cm) by employing a motor crank piston mechanism. Dziekan et al. [26] have developed an economically viable CPR device incorporating a dual-slider crank piston mechanism, which fulfills the requirements of compression frequency (120/min) and compression depth (5 cm), encircling the patient using a pulley belt system. Sadani et al. [27] designed a CPR device with a double-acting pneumatic cylinder piston mechanism, capable of delivering 60 compressions per minute. Hague et al. [28] have developed an affordable CPR device that provides compression for both elderly individuals and infants through the employment of a linear channel. Keakultanes et al. [29] developed a CPR device that can compress the chest at a rate of 80 to 100 compressions per minute and a depth range of 4 to 5 cm using a linear motor. Wei-Sung et al. [30] developed an adjustable piston-based CPR device with a slider crank mechanism that can be used for adults, infants, and pregnant women. Castillo et al. [31] developed a dual-cylinder adjustable piston CPR device and compared it to the Thumper device in terms of weight, setup time, and applied pressure. Garcia et al. [32] compared pressure plots of LUCAS II and Corpuls instruments with screw-piston technology to natural pressure values. Eichhorn et al. [33] compared LUCAS II and Corpuls devices for blood flow and pressure in a mechanical chest model. Xingyuan et al. [34] developed a mechanical CPR prototype with a pneumatic cylinder piston mechanism to mathematically and experimentally validate the superiority of mechanical wave models over manual wave models, besides achieving stable compressions. Al-Mutairi and Al-Aubidy [35] designed a low-cost portable prototype that meets the needs of CPR, ventilation, and real-time monitoring of patient heartbeats. Alam et al. [36] developed a device capable of providing compressions at a rate of 100–120 per minute with a depth of 2 inches, utilizing a motor crank piston mechanism powered by a high-torque gear DC motor.

1.2. Content and contributions

The contributions of our work can be summarized as follows:

- The patented double-crank piston mechanism of the automatic CPR device developed in this study distinguishes itself from conventional piston mechanisms utilized in previously developed devices. Besides achieving standard parameters of CPR, the device with the novel piston mechanism provides contributions such as a compact design, and adjustable and more consistent compression depth.
- An automatic multi-bolt bayonet lock mechanism with radial and axial locking functions has been developed so that the developed device can be quickly installed and started compression during cardiac arrest. With this locking system, the CPR device can be seamlessly integrated into the backplate without requiring any alignment. This automatic locking feature, which does not require manual alignment, provides convenience to the user. In addition, the body designs of the lock system were made in multi-pieces to facilitate manufacturing and low cost, and the production was carried out in this direction.
- The production of complex designs has been accomplished at an industrial scale and with cost-effectiveness, utilizing a desktop 3D printer instead of traditional machining methods. This approach not only facilitated the iterative production process until optimal designs were attained but also held potential benefits for low-volume manufacturing scenarios.

The remainder of this paper proceeds as follows: Section II consists of the materials and methods. The fabrication of the CardiS device is given in Section III. Tests and results of comparing the CardiS with the LUCAS II device are presented in Section IV. Finally, discussion and conclusions and possible future research directions are given in Sections V and VI, respectively.

2. Materials and methods

2.1. Structural design of CardiS device

The developed CPR device is a device that works with a 180-degree phase angle and applies compression to the sternum with a single external piston triggered by a double-crank, dynamic conrod, and two internal pistons. This system is based on Cardiac Pump Theory as it compresses the heart directly between the sternum and spine. Fig. 1 (a) illustrates the double slider crank mechanism design with two internal pistons. To achieve a compact design, a 1:1 crank/conrod ratio was chosen. However, with a 1:1 crank/conrod ratio, a complete cycle of a slider crank is not possible along with a conrod. To achieve a full slider-crank cycle, the conrod length should be more than the radius of the crank during the decompression (non-compression phase). To facilitate this, the conrod has been made dynamic using a compression spring (dynamic conrod spring) that allows the extension of the conrod during decompression. Fig. 1 (b) illustrates this mechanism integrated into the main piston. The main piston has two sections, the main body and two decompression connectors. Decompression connectors connect the main piston body to four decompression springs as shown in Fig. 1 (c), and the main piston upstroke is provided by these decompression springs. The design shown in Fig. 1 (c) is referred to as the compression module. This module has a power transmission mechanism consisting of a DC motor, a flywheel, and an electromagnetic clutch-gear unit. The reason why a flywheel is used in the system is its ability to provide a high mass effect when used in press mechanisms of a small high-rpm flywheel. While the energy required for compressions is provided by this flywheel, the energy of the flywheel is supplied by a brushless DC motor. An electromagnetic clutch is integrated into the flywheel so that compressions can be started and stopped suddenly. High inertia from a small-mass flywheel is achieved by connecting the electromagnetic clutch to a gearbox. The chassis, on the other hand, is designed to mount the power transmission components, piston, horizontal arms, cover, battery, and other parts. The chassis serves as a carrier for these parts and provides resistance against the pressure applied to the chest during the cardiac massage.

Fig. 2 shows the general design of the CardiS device including the compression module, static horizontal arms, telescopic vertical arm, bayonet lock, and backplate. The telescopic vertical arm consists of a vertical arm upper part, a vertical arm lower part, and a split bushing. The upper part of the vertical arm moves within the steel split bushing embedded in the lower part of the vertical arm, allowing users to adjust the height of the compression module according to the patient's chest depth. The height level of the vertical movement is limited by the contact between the stopper on the upper part of the vertical arm and the lower surface of the split bushing. The reason for putting a steel bushing between the vertical arm upper part and the vertical arm lower part is to minimize the stretching that occurs in the telescopic vertical arm during compression. The compression module can slide on the horizontal arms via two mechanical connectors and rotate around the vertical arm to adjust the axial and radial position of the piston on the patient's chest. Also, the vertical arm has two locking units. The first one interlocks the telescopic movement of the vertical arm, the second one provides a tight connection between the vertical arm and the backplate [37].

2.2. Structural design of CardiS lock mechanism

An automatic multi-bolt bayonet lock mechanism with radial and

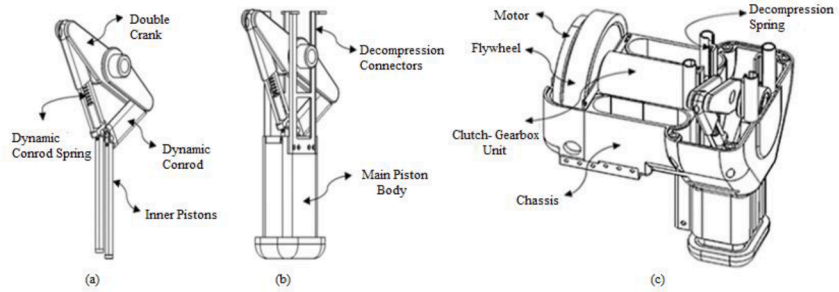


Fig. 1. Double slider-crank mechanism without (a) and with (b) main piston and compression module (c).

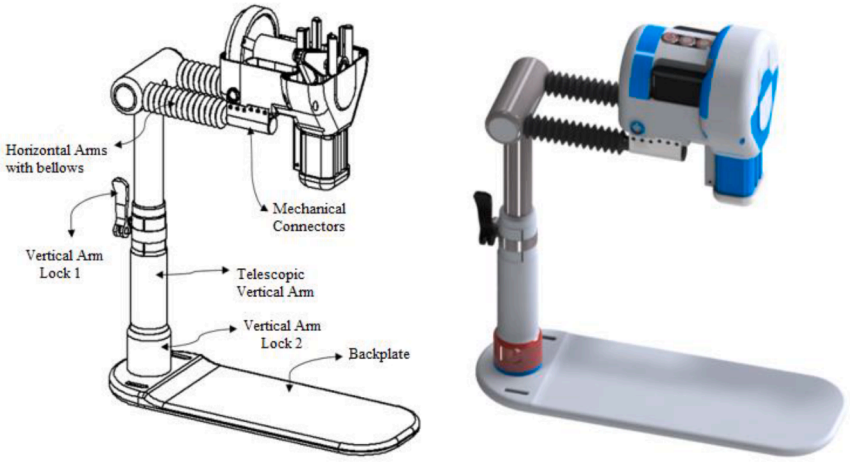


Fig. 2. General design of the CardiS device.

axial locking functions has been designed so that the developed CPR device can be attached to the backplate quickly and effortlessly without any alignment. In advanced designs of bayonet lock mechanisms, semi-automatic bayonet lock systems with manual angular alignment are used. In less sophisticated bayonet lock mechanisms, however, radial and axial locking cannot be achieved with a single mechanism. This is because a separate spring is used for each lock bolt in the lock mechanism.

The bayonet locking mechanism utilized in the Corpuls device, manufactured by GS Elektromedizinische Geräte G. Stemple, represents the utmost advancement known in this realm of technology. This mechanism embodies a semi-automatic configuration, necessitating manual angular alignment. The reason for this is that the correct number of lock bolts is not used for the diameter, depth, and lock slide angle used in the device's design or a design is not made in which the correct number of lock bolts can be used.

In addition, the production of the current design of the bayonet lock system of the device in question involves high costs and technical

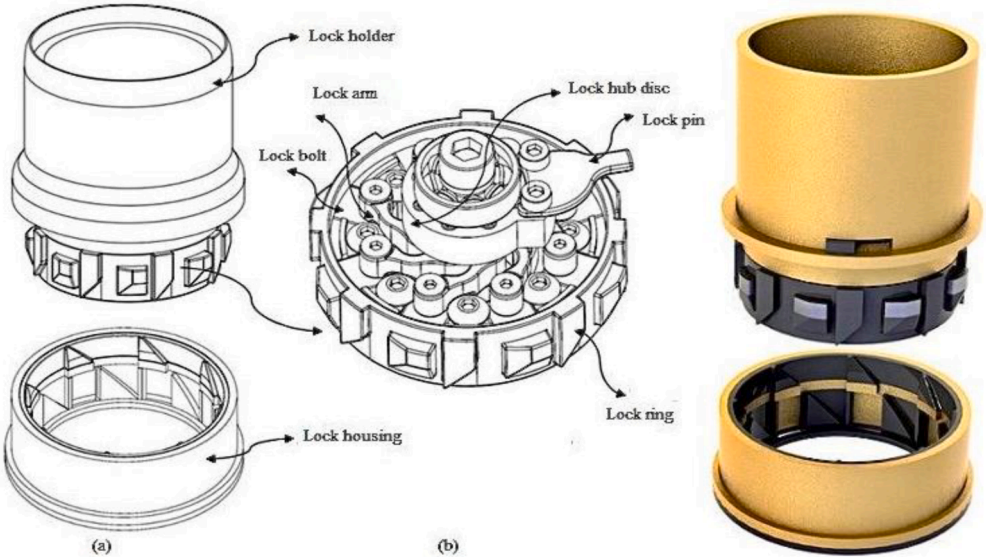


Fig. 3. Lock mechanism (a) and internal mechanism (b) of the complete design.

difficulties with both machining and precision casting methods. The reason for this, then, is their monolithic body designs. The CardiS locking mechanism, which provides enhanced precision in lock slide and alignment through its multi-part body design, can be manufactured with relative ease and at a low cost from a technical standpoint. The complete design of the lock mechanism is illustrated in Fig. 3 (a), while the internal mechanism is depicted in Fig. 3 (b).

In the developed locking system, the strain of the multi-bolt hub mechanism contained within the locked arm is ensured by a single radial spring. This spring facilitates the outward displacement of the lock bolts by the lock arms, thereby ensuring their retention in the open position. The angled and specially designed surfaces at the tips of the lock bolts assist in the inward movement of the lock bolts under pressure. The inward movement distance is defined as the lock distance. The lock bolts are precisely aligned with the locking slots thanks to the delicate designs on the lock ring, and they are securely engaged into the slots through the tension of the spring, ensuring smooth locking. The unlocking of the lock, however, is achieved by manually turning the lock holder, which turns the lock pin, in a counterclockwise direction with a small angle. The rotation direction of the lock holder can be changed by changing the winding direction of the lock spring. In addition, the CardiS lock mechanism has a compact structure thanks to the special geometry of the lock arms.

2.3. Instantaneous compression depth of the system

The free-body diagram of the piston (P), crank radius (R), and conrod length (L) at the start of compression is shown in Fig. 4 [37]. In this diagram, the patient's chest is represented by a mass (m_1), a spring (K_1), and a damper (D_1), while the static structure of the prototype is represented by a mass (m_2), and spring (K_2). Decompression springs (K_3) are not shown in the diagram, but these springs are assumed to be between point O and mass m_1, which is the mass that models the stretching part of the patient's chest. It is a general assumption that the net force along the conrod (L) of the slider crank mechanism is zero, as shown by equal and opposite forces along L in Fig. 5. According to the particular design of the crank-conrod ratio, expressed as 1:1, R is equal to L. It should also be noted that point O shown in the diagram is not fixed in the vertical direction. The maximum piston stroke (S) according to point O is calculated as in Eq. (1), where θ_i represents the crank angle at the start of compression.

$$S = 2R(1 - \cos(\theta_i)) \tag{1}$$

As the instantaneous crank angle (θ) decreases from θ_{max} to 0, the stroke of the piston concerning O is calculated as shown in Eq. (2).

$$x = 2R(cos(\theta) - cos(\theta_i)) \tag{2}$$

The instantaneous crank angle (θ) shown in Eq. (2), and instantaneous crank angular velocity (w) are mathematically expressed as in Eq. (3). For simplicity in piston velocity calculations, instantaneous angular acceleration (α) of the crank pin is assumed to be constant. This assumption will not significantly affect the piston velocity graph, as the instantaneous angular velocity (w) of the crank pin has a higher value than the fluctuation in its value. This assumption is a general case, as in the flywheel calculations of mechanical presses.

$$\theta = \theta_i + w_i t + 0.5\alpha t^2 \qquad w = w_i + \alpha t \tag{3}$$

The instantaneous compression depth (x_1) of the K_1 spring is equal to the difference between the instantaneous piston stroke (x) and the stretching amount of the K_2 spring (x_2) (Eq. (4)).

$$x_1 = x - x_2 \tag{4}$$

According to the free-body diagram shown in Fig. 4, the mathematical model of the general design is obtained as in Eq. (5). F_P and f_s shown in the equation are the force applied by the piston to the spring modeling of the patient's chest and the friction force on the piston, respectively.

$$m_1\ddot{x}_1 + D_1\dot{x}_1 + K_1x_1 = m_2\ddot{x}_2 + K_2x_2 = F_P - (F_{set} + f_s) \tag{5}$$

The instantaneous compression depth (x_1) is obtained as shown in Eq. (6) using Eqs. (2) and (5). To express the equations in a simpler form, M has been written instead of $m_1 + m_2$, and K has been written instead of $K_1 + K_2$.

$$M\ddot{x}_1 + D_1\dot{x}_1 + Kx_1 = 2R[K_2(\cos(\theta) - \cos(\theta_i)) - m_2(w^2\cos(\theta) + \alpha\sin(\theta))] \tag{6}$$

Since the mathematical model contains a quadratic polynomial (θ), the solution of the equation by the method of variation of parameters requires the use of the Fresnel Integral. This requirement does not provide a practical solution. For this reason, the instantaneous angular velocity (w) has been assumed as a constant within the sine and cosine functions and thus the second-order polynomial has been converted into a first-order polynomial. Homogeneous and particular solutions of the general equation of the model, which can be solved by the method of undetermined coefficients with this conversion effect, are shown in Eq. (7) and Eq. (8).

$$x_{1h} = exp\left(-\frac{D_1}{2M}t\right)\left[c_1 cos\frac{\sqrt{4MK - D_1^2}}{2M}t + c_2 sin\frac{\sqrt{4MK - D_1^2}}{2M}t\right] \tag{7}$$

$$x_{1p} = Asin(\theta) + Bcos(\theta) - C + exp(t)[Dcos(\theta) + Esin(\theta)] + exp(2t)[Fsin(\theta) + Gcos(\theta)] \tag{8}$$

The undetermined coefficients (A, B, C, D, E, F, G) obtained from Eq. (8) are respectively shown in Eqs. (9–15).

$$A = \frac{[2RK_2 - 2Rm_2(w_i^2 + 2\alpha w_i + \alpha^2)]D_1\dot{\theta} - 2Rm_2\alpha\left(K - M\dot{\theta}^2\right)}{\left(K - M\dot{\theta}^2\right)^2 + D_1{}^2\dot{\theta}^2} \tag{9}$$

$$B = \frac{A + D_1\dot{\theta}\left[\left(K - M\dot{\theta}^2\right)^2 + 2Rm_2\alpha\left[\left(K - M\dot{\theta}^2\right)^2 + D_1{}^2\dot{\theta}^2\right]\right]}{D_1\dot{\theta}} \tag{10}$$

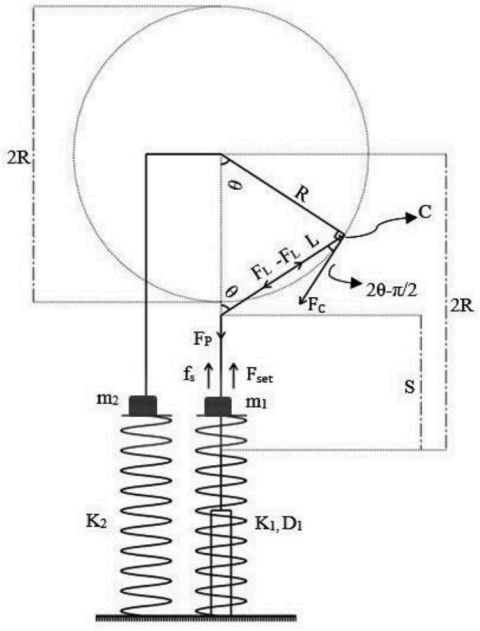


Fig. 4. The free body diagram of general design.

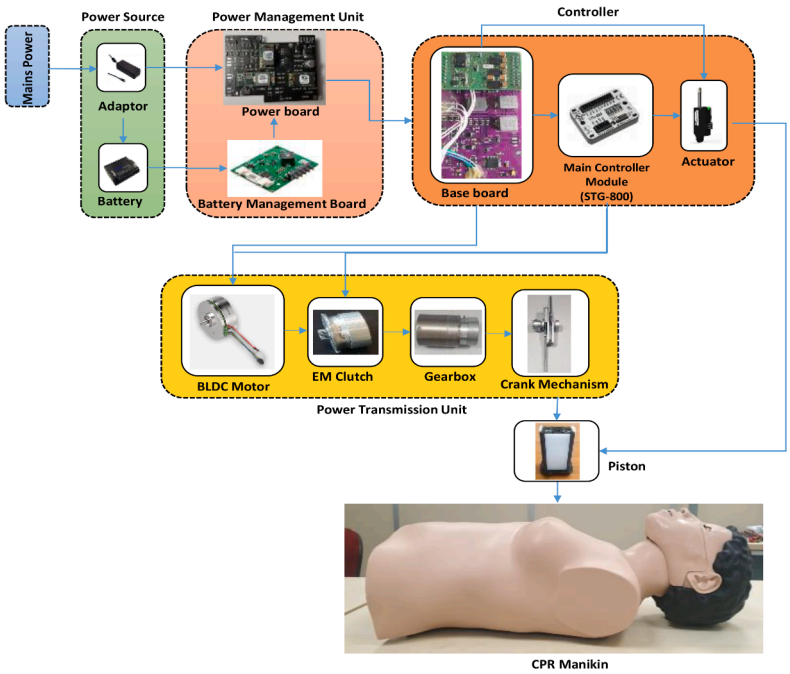


Fig. 5. Electromechanical power flow diagram of the CPR device.

$$C = \frac{2RK_2 cos\theta_i}{K} \tag{11}$$

$$D = \frac{-2Rm_2\left(K + D_1 + M - M\dot{\theta}^2\right)(-2\alpha w_i - 2\alpha^2)}{\left[\left(K + D_1 + M - M\dot{\theta}^2\right)^2 + \left(D_1 + 2M\dot{\theta}\right)\left(D_1\dot{\theta}_t + 2M\dot{\theta}\right)\right]} \tag{12}$$

$$E = \frac{-2Rm_2(-2\alpha w_i - 2\alpha^2)\left(D_1 + 2M\dot{\theta}\right)}{\left[\left(K + D_1 + M - M\dot{\theta}^2\right)^2 + \left(D_1 + 2M\dot{\theta}\right)\left(D_1\dot{\theta} + 2M\dot{\theta}\right)\right]} \tag{13}$$

$$F = \frac{-2Rm_2\alpha^2\left(4M\dot{\theta} + D_1\dot{\theta}\right)}{\left(4M - M\dot{\theta}^2 + 2D_1 + K\right)^2 + \left(4M\dot{\theta} + D_1\dot{\theta}\right)^2} \tag{14}$$

$$G = \frac{-2Rm_2\alpha^2\left(4M - M\dot{\theta_t}^2 + 2D_1 + K\right)}{\left(4M - M\dot{\theta}^2 + 2D_1 + K\right)^2 + \left(4M\dot{\theta} + D_1\dot{\theta}\right)^2} \tag{15}$$

The c_1 and c_2 coefficients obtained when the initial values of the system with $x_1(0) = x_{1i}$ and $\dot{x}_1(0) = V_{1i}$ are substituted in the general solution are shown in Eq. (16) and Eq. (17).

$$c_1 = C - (A + E + F)sin\theta_i - (B + D + G)cos\theta_i \tag{16}$$

$$c_2 = \frac{2MV_i + [C - (A + E + F)\sin\theta_i - (B + D + G)\cos\theta_i]D_1}{\sqrt{4MK - D_1^2}} - \frac{2M[\left(A\dot{\theta}_t + E\dot{\theta}_t + F\dot{\theta}_t + D + 2G\right)\cos\theta_i - (2F + E - B\dot{\theta} - D\dot{\theta} - G\theta_t)\dot{\sin}\theta_i]}{\sqrt{4MK - D_1^2}} \tag{17}$$

$$x_{1g} = x_{1h} + x_{1p} \tag{18}$$

As a result, the equation of the instantaneous compression depth is obtained as in Eq. (18).

2.4. Embedded system design, software, and control

In this study, a brushless DC motor (Maxon EC 90) with hall sensors capable of detecting magnetic field and converting this data into electronic signals to provide information regarding either position or velocity has been utilized. The control of this motor has been provided by a motor driver board (Maxon DEC Module), with a maximum output voltage of 50 V and a maximum output current of 15A. As a main control module, a mini PLC, namely the STG-800, which holds an industrial usage license, has been used. Furthermore, it has been decided to use a position-controlled mini-actuator embedded in the piston bearing to adjust compression depth to a suitable level for the patient's dimensions. For this purpose, a linear actuator (Mighty Zap L7-64PT-3) with a medical-use certificate has been used. This actuator with a stroke value of 27 mm and a force value of 64 N can communicate via TTL/PWM. Two rechargeable RRC-2054 batteries, certified for medical use and international transport certificates, and RRC-PMM240 battery management boards have been used as power sources. With the use of these batteries, it is aimed at providing uninterrupted CPR in cases where access to mains electricity is limited. In addition to these, two PCB boards, a power board, and a base board, have been designed and manufactured to fit the electronic components into the space reserved in the body of the automatic CPR device and to make the electrical connections correctly and regularly.

The main energy source of the developed automatic CPR device is mains power. By converting the mains power to 20 V voltage with the help of an adapter, simultaneously both the battery can be charged and the CPR device can be powered. Unlike other devices, the reason for using two batteries is to make it possible to replace the battery during operation and to provide a longer operating time. The power flow of the adapter and the power card with the battery is controlled by the battery management board. The power supplied to the power board from the adapter or battery is converted to the desired output voltage values (3, 5, 7,12, 20, and 24 V) through DC voltage regulators, and then transferred to the base board. The piston stroke of the device is measured by the TTL pins on the STG-800 and the actuator controlled by the half-duplex circuit on the base board, and thus the compression depth can be precisely adjusted from patient to patient. The speed of the brushless DC motor, on the other hand, is monitored by the PWM signal provided by

the motor driver to the STG-800. When 24 V power is supplied to the electromagnetic clutch from the 24 V regulator located on the power board, the mechanical power of the motor is transferred to the gear. The gear reduces the motor speed by 60:1 and increases the torque 60 times, transferring the mechanical power required for CPR to the crank pin. The crank mechanism, with its S-shape structure, allows the piston to make two strokes in one cycle. Thus, a compression frequency is obtained as 1/30 of the motor speed. The electromechanical power flow diagram of the automatic CPR device developed in line with the above explanations is shown in Fig. 5.

The software of this study has been carried out using the STM32F091, an ARM-based microcontroller, on the STM32CubeIDE platform using the C language. The PWM signals have been generated to drive the motor with STG-800 and motor speed control has been provided with it. Hall sensors have been used to use the speed of the motor as a feedback signal. By comparing the PWM signal with the signal obtained from the hall sensors, necessary arrangements have been made on the data obtained from this comparison so that the motor can give the desired compression value. The actuator, on the other hand, has been controlled by TTL communication. Position data is calculated by the microcontroller to an analog voltage from a linear potentiometer mechanism embedded in the vertical arm. This obtained data is then sent to the actuator that has a built-in controller unit. Thus, the compression depth can be precisely adjusted within the range of 5 to 6 cm.

3. Fabrication of CardiS device

The fabrication processes of the CPR device will be explained through three primary categories: Fabrication of compression module, fabrication of horizontal arms and telescopic extension mechanism, and fabrication of backplate and bayonet lock system.

3.1. Fabrication of compression module

The flywheel, one of the most critical components of the compression module, has been manufactured in stainless steel. To integrate the produced flywheel into the DC motor, the motor's rotor has been removed with a disassembly tool. The flywheel has been mounted on the removed motor rotor with a press fit. A 4 mm centering shaft with a small bearing has been mounted to the motor shaft to center the motor-flywheel unit and the electromagnetic clutch. To integrate the armature disc of the electromagnetic clutch providing a start-stop function for chest compression, into the flywheel, on the other hand, a clutch connector has been designed and produced from stainless steel by laser cutting technique. The clutch disc, on the other hand, has been assembled to the clutch connector using aluminum rivets.

To boost the reflected inertia of the flywheel to the crank, a 60:1 gear and gearbox have been used between the electromagnetic clutch and the crank. The integration of the rotor of the clutch and the gearbox has been ensured by a shaft placed with two ball bearings between the input gear of the gearbox and the rotor of the electromagnetic clutch. The stator of the clutch has been, on the other hand, integrated into the gearbox body via laser weld. The gear assembly has been completed by a bearing unit integration on the output of the gearbox by four screws.

The crank mechanism and gearbox motion coupling, necessary to convert the circular motion into vertical motion, have been provided by a flat wedge mounted on the output shaft of the gearbox. Steel arms of the double crank have been prepared by precision cutting i.e., wire-cut and grinding processes, and then these crank arms and pins have been combined with argon welding to form a monolithic double crank part. This double crank mechanism is designed to operate with a 180-degree phase difference so that two compression motions can be achieved within the 360-degree motion of the crank.

One of the two arms of the double crank moves downward, applying compression to the patient for heart massage, while the other performs an upward retraction movement. This double crank mechanism receives support from bearings connected to the chassis with two screws. Dynamic conrod mechanism, on the other hand, consists of conrod upper part, conrod lower part, conrod pins, internal pistons, internal piston bearings, and spring. Conrod arms transmit the force they receive from the crank pin to the internal pistons, and thanks to the spring and slider system they contain, they have the feature of being shortened during compression and extended during retraction. On the other hand, internal pistons transmit the force they receive from the conrod lower parts to the piston, allowing the piston to perform the necessary compression for cardiac massage. To achieve the double crank-dynamic conrod motion coupling, first of all, the crankpins have been inserted into the holes of the conrod arms and the conrod arms have been articulated with the shaft mounting to the double crank. Later, internal piston bearings were produced so that the internal pistons of the conrod could hold onto the piston. Thus, the double crank-dynamic conrod integration has been completed successfully. All internal mechanisms from the motor to the double crank-dynamic conrod are shown in Fig. 6.

The piston, which performs the task of transmitting the force received from the internal pistons to the patient's chest, has been produced from PC/ABS FR thermoplastic material with a 3D printer. TIVAR 1000 films with a quite low friction coefficient and wear have been placed in the slots on the side surfaces of the piston with some tolerance so that the piston could move easily. A disposable silicone piston head that can be changed from patient to patient has been added to the piston. The fact that the piston head can be removed and attached serves an important purpose such as hygiene, while the fact that the head is made of silicone material, which is a natural elastomer, is very important in terms of providing a soft surface during cardiac massage. In addition, since the piston upper part, which was first produced using a monolithic aluminum material, was found to have problems such as insufficient compressive strength and wear, the production of the new piston upper part was carried out using multi-piece and steel material. The chassis, which acts as the carrier of parts such as internal mechanism parts, batteries, and electronic components, and provides mechanical stability during CPR has been manufactured from PC/ABS FR material. The cover, on the other hand, designed to be compatible with the chassis and manufactured from the same material, serves to protect the internal mechanisms of the device against external factors. The horizontal arm-chassis connectors have been made of stainless steel before, and milling processes, drilling processes, and countersinking processes have been applied for horizontal and vertical screw holes. However, low cost and easy manufacturability are one of the main goals of the device. Therefore, the horizontal arm-chassis connection parts have also been manufactured with the PC/ABS FR material, which is used for producing larger-sized components like the chassis and cover. All components of the produced compression module are shown in Fig. 7.

3.2. Fabrication of horizontal arms and telescopic extension mechanism

The horizontal arms have been manufactured of stainless steel for high flexural stability against the cantilever force applied by the patient's chest. The horizontal arm-vertical arm connector, on the other hand, which is mounted to the horizontal arms by tight fit and to the vertical arm by threading, has been produced from 5000 series aluminum material with a diameter of 5 cm. The reason the part has been produced on such a large diameter was that it was one of the critical areas of the device that could flex. However, it was concluded that it would be better aesthetically to produce this piece on a slightly smaller scale compared to the dimensional compatibility of the other components of the device with the device. Therefore, this part has been re-produced with a diameter of 4.2 cm with 7000 series aluminum material instead of 5000 series by taking into account the flexing factor. With this improvement, the weight of the part decreased from 520 g to 310 g, contributing 210 g to the weight of the entire device, and a more aesthetic appearance has been achieved. Finally, a hard anodizing

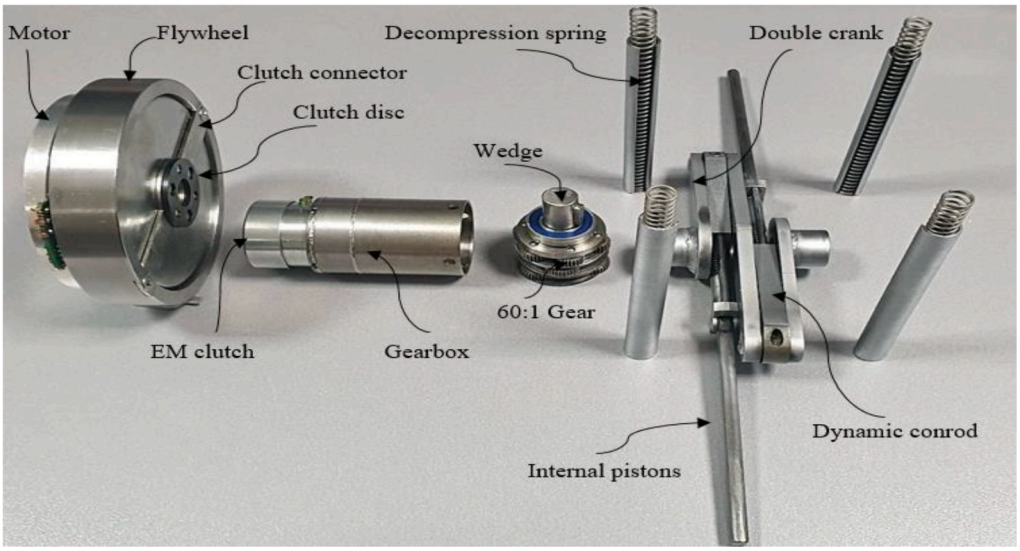


Fig. 6. Internal mechanism of the compression module.

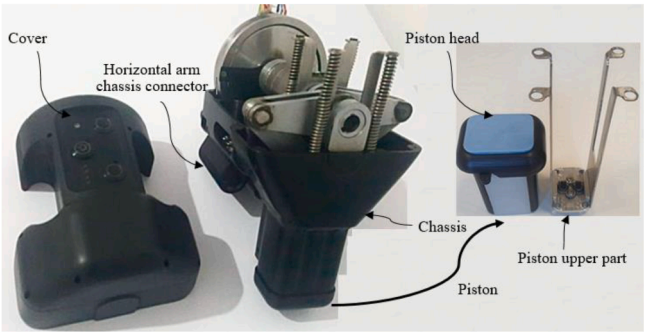


Fig. 7. Compression module.



Fig. 8. Horizontal arms, horizontal arm vertical arm connector, and telescopic extension mechanism components (a), with the telescopic vertical arm closed (b), with the telescopic vertical arm open (c).

process has been applied to the manufactured connector and thus hardness, high durability, and smoothness. Additionally, it has prevented issues such as scratching and corrosion. In addition to these, stopper parts designed to prevent the compression module from slipping off the horizontal arms and the plugs designed for the side and body holes in the connector have been manufactured from PC/ABS FR material.

The telescopic extension mechanism consists of a vertical arm upper part, a vertical arm lower part, and a split bushing. The upper part produced from a 6063 series aluminum tube, moves in a split bushing embedded in the lower part, allowing manual adjustment of the right-left and up-down position of the horizontal arms on the patient. The height level of vertical movement is limited by the contact of the stopper on the upper part with the lower surface of the split bushing. The reason for using a steel split bushing between the upper part and the lower part is to enhance the locking function of the vertical arm with a mandrel locking unit. Furthermore, with the use of this bushing, the diameter of the telescopic extension mechanism has been reduced to be suitable for the grip comfort of a small hand. Due to a high strength requirement, the lower part, on the other hand, has been produced with PA66GF30 (30 % glass fiber reinforced polyamide 6.6) material by machining. An off-the-shelf mandrel locking mechanism has been used to keep the telescopic extension mechanism at the desired height. The mandrel is used with a collet to clamp the vertical arm. This collet has been produced from stainless steel with laser cutting technique and bending. Horizontal arms, horizontal arm-vertical arm connector, and telescopic extension mechanism components and the assembly of these components are shown in Fig. 8.

3.3. Fabrication of backplate and bayonet lock system

The back support required for compressions is provided with a backplate. The production of the backplate, just like the vertical arm lower part, has been carried out by machining techniques from PA66GF30 material. For the backplate, which weighs about 1200 g, to be mounted smoothly to the bayonet lock mechanism, various operations such as opening a connection hole, tapping, and invoice opening have been applied to the backplate. In addition, suitable belt slots have been created to fix the backplate to the stretcher with a belt.

The upper part of the lock system, developed to connect the backplate and the vertical arm, consists of four parts: the shouldered part, the base part, a lock ring, and the lock holder. The shouldered part is made of aluminum and the base part is made of steel, and a hard anodizing has been applied to both parts. Additionally, when the lower lock was mounted and tested manually, it was seen that the lock system was not radially stable enough. To solve this problem, a facing ring was produced by laser cutting technique and this ring was processed monolithic to the shouldered part. Eight U profiles were opened on the side parts of the base part, on the other hand, and this base part was welded to the shouldered part from the inside and outside by laser welding. The lock ring designed to align the lock and the lock holder designed to separate the upright handle from the backplate were produced from PC/ABS FR material by 3D printing technology, as they do not require strength.

The bayonet lock has upper and lower parts to be mounted to the

vertical arm and backplate. The lower part consists of an aluminum main body and two collets enabling radial alignment for locking operation. The main body of the lower part has connection holes to be screwed to the backplate and shouldered slots functioning as support for axial loads. The collets produced from PC/ABS FR material are glued to the main body with epoxy adhesive.

The internal mechanism of the lock system consists of lock arms, lock bolts, a lock pin, a radial spring, a lock hub disc, lock arm bolts, a lock hub bolt, and two bearings. Since the bayonet lock is designed to be compact and has to work with low tolerances the lock arms, the lock hub disc, and the lock pin have been manufactured with PC/ABS FR material. On the other hand, the lock bolts have been manufactured in steel and PC/ABS FR materials. All components of the bayonet lock system are shown in Fig. 9.

The production of the bayonet lock system developed to connect the backplate and the upright arm, which provides the necessary back support for compressions, has been carried out easily and at a quietly low cost thanks to its multi-part design. Thanks to this lock system, which does not require any alignment and has an automatic locking feature with a lower force, fast installation, which is a crucial parameter for automatic CPR devices, can be easily achieved. The general assembly of the CardiS device consisting of the compression module, horizontal arm, telescopic extension mechanism, backplate, and bayonet system is shown in Fig. 10.



Fig. 10. The general assembly of the CardiS device.

4. Tests and results

4.1. Comparative analysis of CardiS and LUCAS II devices regarding standard CPR parameters

This section of the experimental study involves comparing the developed CPR device with the LUCAS II device over standard CPR parameters. This comparison has been carried out with a special CPR

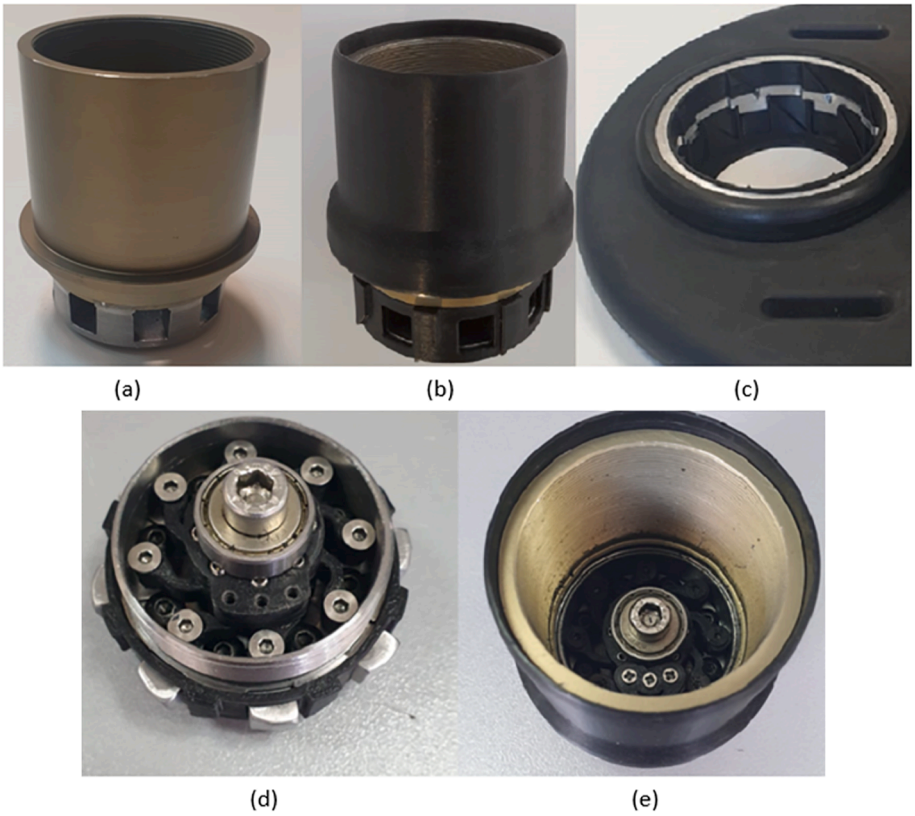


Fig. 9. The shouldered and base part of the lock system (a), the general assembly of the upper part (b), the lower part(c), the internal mechanism (d), and mounted on the bayonet lock (e).

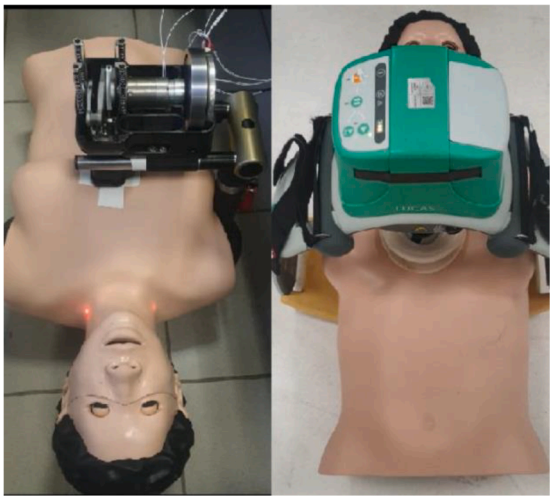


Fig. 11. Standard CPR tests of the CardiS and the LUCAS II devices.

manikin called the CPR Lilly Pro+ (Fig. 11). The utilized CPR manikin enables the measurement of critical parameters such as chest compression rate, compression depth, the amount of chest recoil during compression, and the compression fraction, which is the ratio of the time spent on compression to the operation time. According to the AHA guidelines, for an ideal CPR, it is necessary to perform compressions between 100 and 120 per minute, ensure a compression depth between 5 and 6 cm, and achieve full chest recoil, defined as returning the patient's chest to a neutral position. Additionally, the chest compression fraction should be at least 80 %.

The standard CPR tests of the CardiS and the LUCAS II device have been conducted over 1 min, 5 min, and 10 min. These tests have been repeated ten times for each duration. The results obtained during these test durations are presented in Table 1. The CPR manikin used in the testing process has been fixed to the backplate of both devices. As per the findings, the recoil value represents an offset value of compression depth, stemming from the preload effect during device installation on the manikin. The actual compression depth is the difference between the compression depth obtained from the manikin and the recoil values. The prerequisite for achieving full recoil is that the recoil value should never exceed the initial value. The obtained data also corroborates this condition. Any decrease in the recoil value below the initial value, on the other hand, is attributed to the slight movement of the manikin. Therefore, both the LUCAS II and CardiS devices demonstrate full recoil capacity based on the results obtained. The LUCAS II is capable of performing 100 compressions per minute. To enable an objective performance comparison among the devices, the CardiS has also been adjusted to apply compressions of 100 per minute. The compression fraction parameter, on the other hand, differs between manual and mechanical chest compressions. Manual chest compressions typically demonstrate a lower chest compression fraction compared to mechanical chest compressions, primarily due to factors such as fatigue, insufficient depth, or rescuers' interchange to take over CPR. Both compared mechanical chest compression devices offer two options for compression rhythm: 30:2 and uninterrupted compression. In the 30:2 option, there is a pause in compression for 1.5 s after every 30 compressions for two consecutive ventilations. In the conducted tests, the uninterrupted compression option was chosen, so there was no need to measure compression fraction. However, if the 30:2 option had been used, pauses between compressions would have required measurement of the compression fraction.

Table 1
CPR test results of LUCAS II and CardiS.

		LUCAS II		CardiS	
Time(min)		Mean	Std. Dev.	Mean	Std. Dev.
Compression Freq. (/min)	1	100.30	0.4830	100.20	0.4216
	5	100.22	0.2201	100.18	0.1475
	10	100.18	0.1549	100.17	0.1160
Compression Depth (cm)	1	5.44	0.0776	5.18	0.0419
	5	5.43	0.0791	5.19	0.0442
	10	5.40	0.0801	5.17	0.0487
Recoil (cm)	1	0.24	0.0200	0.13	0.0219
	5	0.23	0.0216	0.13	0.0220
	10	0.24	0.0229	0.14	0.0267

Furthermore, although both devices provide the compression depth range, it is seen that the developed CardiS device provides a more consistent depth according to the standard deviation results obtained as a result of ten tests. The results of the 1 min, 5 min, and 10 min compression depth tests conducted on the manikin are illustrated in Fig. 12, Fig. 13, and Fig. 14 respectively. As a result, in this experimental study, both the LUCAS II and the CardiS device have successfully fulfilled the conditions specified in the AHA guidelines. Video materials showing the CardiS CPR device works in unloaded conditions and on a CPR manikin, and pictures of some of its fabricated parts are available in the Supplementary Materials section.

4.2. Comparison of compression and decompression timings of CardiS and LUCAS II devices

The MATLAB platform is utilized for the computation of the mathematical model solution for compression depth. Table 2 shows the parameters established to attain 50 mm compression depth. For compact dimensions and weight considerations, the flywheel inertia is constrained to 0.0011 kg.m^2. Setting a compression frequency of 100/min, the initial speed of the flywheel and motor is calculated as 3380 rpm to align with this frequency. The crank angle at the start of compression is determined as 1.07 rad. To facilitate piston recoil, four decompression springs with a stiffness of 383 N/m are employed. The model accounts for friction between the piston and piston bearing, with a friction coefficient specified as 0.1 based on Tivar 1000 material data. The patient's chest is represented as a linear compression spring characterized by a stiffness of 10,000 N/m, a damping coefficient of 250 Ns/m, and a dynamic mass of 1 kg. This representation is based on previous studies [38,39]. The main piston mass (0.1 kg) is combined with the patient's chest moving mass, denoted as m_1. The miniature electromagnetic clutch interposed between the flywheel-motor unit and the gearbox has a nominal torque of 1.5 N.m. All parameters, excluding those about the patient model as outlined in Table 2, have been incorporated into the practical implementation tests. The current values of patient chest model parameters for the CPR manikin are unknown. Nevertheless, it is assumed that the K_1 value is close to 10,000, and the D_1 and m_1 values

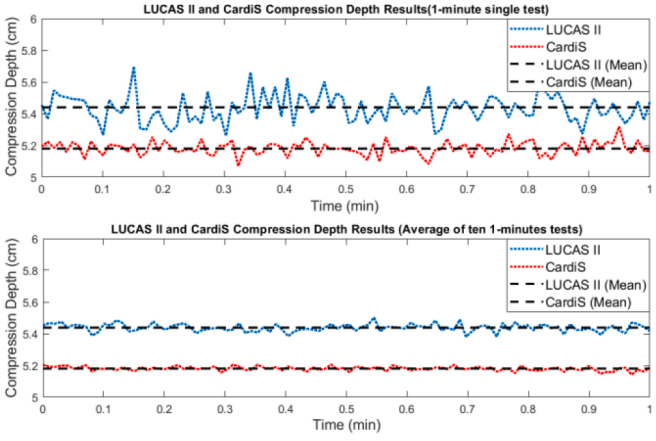


Fig. 12. 1-minute compression depth results of LUCAS II and CardiS.

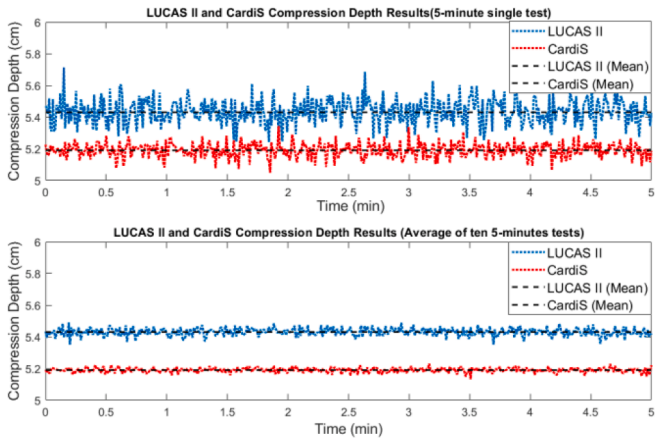


Fig. 13. 5-minute compression depth results of LUCAS II and CardiS.

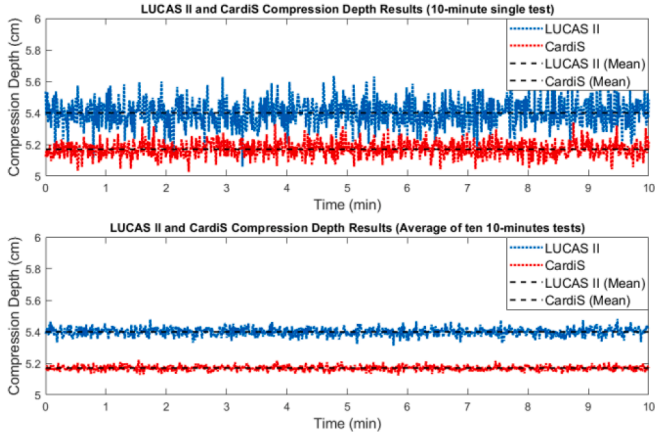


Fig. 14. 10-minute compression depth results of LUCAS II and CardiS.

Table 2
Parameters in mathematical modeling for 50 mm compression.

R: 0.059 m	K_2: 75000 N/m	N_i: 3380 rpm
L: 0.059 m	K_3: 1532 N/m	θ_i: −1.07 rad
I_{FI}: 0.0011 kg.m^2	m_1: 1 + 0.1 kg	f: 100/min
τ_{CI}: 1.5 Nm	m_2: 2.2 kg	F_{set}: 26.6 N
D_1 : 250 N.s/m	n: 60:1	E_G: 0.90
K_1: 10000 N/m	x_{1i} : 0	k_s: 0.1

are below the indicated values in the table.

The comparison of the compression and decompression timings obtained from the average of the one hundred compressions using the video analysis method for CardiS and LUCAS II devices is shown in Fig. 15. Due to the inability to adjust the frequency in the LUCAS II device, to enable an objective comparison, the frequency for CardiS has been standardized at 100/min in both its mathematical model and practical implementation tests, and the results have been evaluated based on this frequency. Graphical analysis reveals that the mathematical outputs of the CardiS show similarities with the actual results observed on the CPR manikin. Small timing differences arise from the unknown manikin model parameters and differentiating frictions in the CardiS device. In the obtaining of the graphs, the compression depth and compression period obtained from the manikin and the timing values obtained from the video analysis method have been implemented on the known compression waveforms of LUCAS II and CardiS devices. These compression waveforms are trapezoidal for LUCAS II and sinusoidal for CardiS [33,37]. When the results of CardiS and LUCAS II devices were compared it was observed that the compression and decompression timing of the LUCAS II device has been substantially different from the timing of CardiS. According to the results obtained, CardiS reaches 50 mm compression depth at an average of 199.6 ms, while LUCAS II reaches the same depth at an average of 300,3 ms. Furthermore, while the time for the LUCAS II device to reach an uninterrupted compression of approximately 4.5 cm is 159.2 ms, CardiS can reach the same depth at 130.4 ms. For the LUCAS II device piston velocity drops to zero for a short instant at about 4.5 cm compression depth which seems as a short interruption in compression. In the study conducted by Eichorn et al. [33], the uninterrupted compression time of approximately 4.5 cm of the Corpuls device has been measured to be 20 ms shorter than the LUCAS II device, and the maximum compression time has been 20 ms longer. In the mentioned study, it has been shown that a 20 ms longer maximum compression time provided better blood pressure for the thoracic pump. Since CardiS is designed according to the cardiac pump theory, it provides shorter compression time or lower duty cycle compared to LUCAS II. According to this theory, higher blood pressure can be achieved with a duty cycle of 30–35 % [40–42].

Additionally, the time interval during which blood returns to the heart within a compression period is the time remaining from the compression. According to the results obtained, the remaining time for the blood to return to the heart within a compression period in the LUCAS II device is approximately 299.7 ms. In the CardiS device, this time has been obtained as approximately 400.4 ms. Since the blood returning to the heart flows slowly, the time allocated for it to return to the heart should be long enough. The fact that this period is longer in the CardiS device compared to LUCAS II may contribute to increasing the amount of blood returning to the heart and achieving shockable cardiac perfusion.

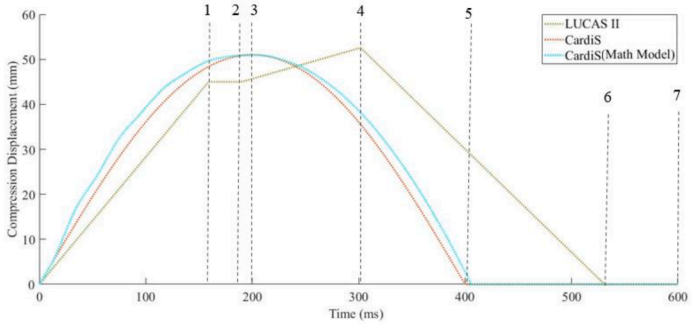


Fig. 15. The comparison of the compression and decompression timings for CardiS and LUCAS II devices. 1. End of uninterrupted compression of LUCAS II, 2. End of interrupted compression of LUCAS II, 3. End of CardiS compression, 4. End of LUCAS II compression, 5. End of CardiS decompression, 6. End of LUCAS II decompression, 7. Start of the next compression of CardiS and LUCAS II.

5. Discussion

The current CPR devices apply the 50 % duty cycle recommendation in the AHA guidelines regarding piston timing to reduce the fatigue factor for standard manual applications. On the other hand, there are many studies stating that the duty cycle should be in the range of 30–35 % to obtain the natural timing and natural flow form of the heart [40–43]. These studies suggest that the use of these duty cycle percentages leads to superior chest compression quality. However, there is no CPR device study developed according to the specified duty cycle range yet. Therefore, the CardiS device, which provides a duty cycle of approximately 33 % thanks to its piston mechanism consisting of a double crank and dynamic conrod, is aimed to fill this gap in the literature.

In blood flow in the normal cardiac cycle, the pressure difference between the left ventricle and the aorta is proportional to the acceleration of blood flow from the left ventricle [44,45]. According to CPR studies on blood flow mechanism, chest compression with high initial acceleration can directly initiate cardiac compression and blood flow close to normal blood flow can be obtained [45]. Concluding compression with a low finish acceleration can prevent life-threatening complications such as lung laceration and liver and pancreas rupture that occur with the use of current devices [46]. Although many mechanical CPR devices developed can reach the manual CPR parameters described in the AHA guidelines, none of them has yet been proven to be superior to manual CPR. The reason a machine could not be superior to a human could be due to initial momentum. The human upper body mass can be a good source of momentum during manual CPR, allowing blood to flow in a pulsatile waveform with high acceleration. In current mechanical CPR technologies, such a momentum source is not available. Obtaining high initial momentum in CPR devices is possible with the high mass effect on the piston and the high initial speed of the piston. A high mass effect can be generated with high rotational inertia. This inertia might be provided by the power transmission unit consisting of a flywheel, electromagnetic clutch, and gear unit in the structure of the CardiS device. The high initial speed of the piston is, on the other hand, related to the structure of the slider crank mechanism, and a crank angle range must be selected for this. Although a crank angle value of approximately 60 degrees is quite practical for dynamic force evaluations, the duty cycle at this degree is 16.6 %. This value is too low for effective chest compression. The reason for using a double crank instead of a single crank in the piston mechanism of the CardiS device is to reach the recommended duty cycle range by doubling this duty cycle value. It is predicted that the CardiS device, which has been shown to successfully provide standard CPR parameters in this study, can also going to provide high initial acceleration thanks to this special piston mechanism. To verify this prediction, a chest compression blood flow testing mechanism needs to be established based on a determined blood flow movement model.

6. Conclusions and future work

In this study, a portable, small, and compact mechanical CPR device with a special piston mechanism consisting of a double slider crank and dynamic conrod, has been designed and manufactured. The patented piston mechanism of the automated CPR device developed is different from the piston mechanisms developed so far. Complex designs have been manufactured at a quietly affordable cost with a 3D printer instead of machining. These designs have been manufactured with PC/ABS FR, a thermoplastic material that can withstand high temperatures and has very high strength, used in many sectors, especially automotive and medical. Furthermore, an automatic bayonet lock system with radial and axial locking functions has been developed so that the developed device can be integrated into the patient quickly and effortlessly during cardiac arrest. Thanks to this lock system, quick installation can be carried out with less force without any alignment.

The developed CardiS device has been compared with the LUCAS II device over standard CPR parameters using a special CPR manikin called CPR Lilly Pro + . The standard CPR tests have been conducted over durations of 1 min, 5 min, and 10 min. Since LUCAS II can perform 100 compressions per minute, CardiS has also been adjusted to perform 100 compressions per minute to make an objective performance comparison between devices. According to the results obtained, both devices have shown full recoil capacity. In addition, although both devices provide a compression depth range, it has been observed that the developed CardiS device provides a more consistent depth. Since uninterrupted compression was preferred in the tests, there has been no need to measure the compression fraction. In conclusion, it can be seen that both devices successfully meet the standard CPR conditions specified in the AHA guidelines, but the CardiS provides a more consistent compression depth.

Compression and decompression timings obtained from the average of one hundred compressions have been compared using the video analysis method for CardiS and LUCAS II devices. Graphical results have been obtained by applying the compression depth and compression time obtained from the manikin and the timing values obtained from the video analysis method on the known compression waveforms of LUCAS II and CardiS devices. When the results were compared, it was seen that the CardiS device provided shorter compression time or lower duty cycle compared to LUCAS II. In addition, it has been determined that the time allocated for blood to return to the heart is longer in the CardiS device. This may allow the CardiS device to provide a more effective compression application process and perform heart massage more efficiently.

In future studies, the foremost objective entails the development of a test tool for acceleration measurement to compare the initial accelerations between the CardiS and LUCAS II devices. This comparative analysis seeks to elucidate the efficacy of the CardiS device in implementing the cardiac pump theory vis-à-vis the LUCAS II device. Additionally, a novel blood flow test mechanism will be developed to gauge the quantity of blood propelled from the heart to vital organs during compression and the amount of blood flow achieved by the proposed CPR device on this test mechanism will be compared with the LUCAS II device. By evaluating the results obtained from the blood flow test mechanism, it will be tried to prove the superiority of blood circulation provided by the CardiS device over the LUCAS II device.

6.1. Limitations

Due to the long and complex numerical methods required for solving the mathematical model of the developed device, some parameters have undergone mathematical approximation and physical assumptions. Additionally, although the compression depth and recoil parameters are displayed in real-time on the interface of the CPR manikin's application called CPR Lilly, the obtained data is not automatically presented in a data file format. Therefore, manual recording of the data has been necessary for each compression, significantly limiting the efficiency of the data collection and analysis processes in the experimental study. In addition, the current values of the patient chest model parameters of the CPR manikin (K_1, D_1, and m_1) are unknown. Finally, there are very few studies on compression timing in the literature. Therefore, comprehensive comparative studies on different devices and methodologies are needed.

CRediT authorship contribution statement

Ahmet Kağızman: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Volkan Sezer:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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